

Task 5 Exploratory Data Analysis (EDA) Report of Findings

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Topic: Exploratory Data Analysis (EDA) Production Pipeline

Dataset Name: Palmer Penguins Morphological Specimen Audit

1 .Executive summary

This report details the visual and statistical analysis of morphometric parameters across isolated avian study specimens. The analytics framework checked the underlying shape profiles, data health anomalies, and directional correlations across numerical attributes. The goal is to provide a clean data foundation for downstream predictive models.

- Data profile: The sanitized base matrix contains structural continuous features across three distinct taxonomic groups.
- Data health: Programmatic cleaning successfully removed 318 missing value vectors in unstructured attributes to preserve calculation integrity.
- Core takeaway: Morphological variations provide highly localized separation zones, proving that classification models can achieve high precision using baseline linear combinations.

2 .Statistical data profiling

An initial assessment evaluated column constraints, structural null distributions, and continuous measure tendencies.

- Class distribution: The target variable exhibits a well-balanced distribution profile. The data contains 44% Adelie, 36% Gentoo, and 20% Chinstrap specimens. This balance ensures that machine learning algorithms will not favor a single dominant group.
- Measurement scale: Summary statistics highlight significant variations in physical metrics across the population. Body mass ranges from 2700 g to 6300 g , while flipper length varies between 172 mm and 231 mm .

3 Univariate feature topologies

Univariate plots analyze data attributes independently to identify distribution patterns and outliers.

Flipper length distribution analysis

- Observation: The flipper length histogram shows a distinct bimodal distribution pattern with major peaks centered at 190 mm and 215 mm.
- Insight: This structural pattern confirms that the population splits into two size groups, which likely corresponds to species-level physical variations.

Target-stratified body mass outlier evaluation

- Observation: The boxplot shows clear differences in median weights between categories. The Gentoo species maintains a significantly higher mass range compared to the other two groups.
- Insight: Data point distribution is clean and shows no extreme anomalies. This uniform variance confirms that the physical data is highly reliable for model training.

4 .Multivariate relationship tracking

Multivariate tracking maps interaction vectors, linear dependencies, and clustering behaviors across continuous metrics.

Numerical correlation topology matrix

- Observation: The masked correlation heatmap reveals a very strong positive relationship (0.87) between Flipper Length and Body Mass. Conversely, Culmen Depth correlates negatively with both flipper length (-0.58) and body mass (-0.47).
- Insight: Flipper length scales proportionally with overall skeletal mass. Culmen depth trends in the opposite direction, serving as a distinct physical differentiator.

Spatial feature clustering grid

- Observation: Cross-variable scatter plots show clear spatial separation zones. While individual metrics overlap across classes, cross-plotting dimensions (such as Culmen Length vs. Culmen Depth) completely isolates the three groups.
- Insight: The features show low mutual overlap, meaning standard linear classifiers can easily separate the groups without requiring complex non-linear models.

5. Master-level interview response manual

1 Which plots do you use to check correlation?

- Heatmap: Summarizes large matrix-wide linear interactions into a single compact grid using color-coded numerical correlation scores.
- Scatterplot: Displays localized density distributions, directional paths, and linear relationships between two variables.

2 How do you handle skewed data?

- Log transform: We apply $\text{np.log1p}()$ to compress long right-hand tails.
- Power transform: We use Box-Cox or Yeo-Johnson algorithms to equalize variance and guide data toward a symmetric normal distribution profile.

3 How to detect multicollinearity?

- Correlation score: Scanning the feature matrix for high correlation values that exceed 0.80 .
- VIF computation: Checking the Variance Inflation Factor for each variable. A score above 5.0 indicates redundant variance, making that column a candidate for removal.

4 What are univariate, bivariate, and multivariate analyses?

- Univariate: Evaluating a singular attribute profile (e.g., using histograms) to isolate individual frequency counts and skewness.
- Bivariate: Comparing two distinct parameters (e.g., using scatter plots) to evaluate directional trends.

- Multivariate: Investigating multi-axis relationships simultaneously (e.g., using pairplots) to identify distinct population clusters.

5 Difference between heatmap and pairplot?

- Heatmap: A concise grid that maps matrix-wide correlation coefficients using numerical values and color scales.
- Pairplot: A comprehensive plot matrix showing individual scatter plots and distribution curves for every pair of continuous variables.

6 How do you summarize your insights?

- Business framing: We frame insights around technical drivers and pipeline impacts rather than simply listing basic summary metrics. This includes detailing data cleaning choices and identifying key predictive variables.

7 What is EDA and why is it important?

- Definition: Exploratory Data Analysis is the systematic process of using visual graphics and statistical profiling to audit data integrity.
 - Importance: It reveals data leakage, identifies missing data anomalies, uncovers structural skewness, and ensures your data satisfies mathematical model assumptions.
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